Biology

Time: 3 Hours 15 Minutes **Total Marks: 70**

Section-I (Objective type Question)

The following Question Nos. 1 to 28 there is only one correct answer against each question. For each question, mark the correct option on the answer sheet:

28 x 1 =28

1. Which one is diploid structure?
   (A) Ovaries (B) Endosperm
   (C) Zygote (D) All of these
   ANS:(C) Zygote
2. Oncogene is responsible for-
   (A) Cancer (B) AIDS (C) Tuberculosis (D) Polio
   ANS:(A) Cancer
3. Which one is hydrophytic plant?
   (A) Trapa (B) Opuntia
   (C) Dulbergia (D) Acacia
   ANS:(A) Trapa
4. Fertilization outside the body is called-
   (A) In Vitro (B) In vivo (C) Both (D) None
   ANS:(A) In Vitro
5. World Environment day is celebrated on-
   (A) 5th June (B) 20th December
   (C) 15th March (D) 7th July
   ANS:(A) 5th June
6. Sertolli cells are found in-
   (A) Testis (B) Uterus
   (C) Ovary (D) Liver
   ANS:(A) Testis

7. Operon Model is representation of-
   (A) Gene synthesis (B) Gene expression
   (C) Gene regulation (D) Gene function

   ANS:(D) Gene function

8. Which one is not hermaphrodite animal?
   (A) Tape worm (B) Earth worm
   (C) House fly (D) Leech

   ANS:(C) House fly

9. Which nitrogen base is absent in DNA?
   (A) Thymine (B) Uracil
   (C) Guanine (D) Cytosine

   ANS:(B) Uracil

10. Red Date Book includes list of -
    (A) Endangers Plants (B) Rare Plants
    (C) Threatened Animals (D) All of the above

    ANS:(D) All of the above

11. Thalamus is edible part of-
    (A) Annoma (B) Apple (C) Orange (D) All of these

    ANS:(B) Apple

12. DNA is genetic material of-
    (A) TMV (B) Bacteriophage
    (C) Both (D) None

    ANS:(B) Bacteriophage

13. Population explosion results in-
    (A) Reduction in Income (B) Reduction in land
    (C) Reduction in minerals (D) All of these

    ANS:(D) All of these

14. Acrosome is a part of -
    (A) Head of Human sperm (B) Middle part of human sperm
    (C) Primary oocyte (D) Blastocyst

ANS:(A) Head of Human sperm

15. Which one is a bacterial disease?
    (A) Leprosy (B) Tuberculosis
    (C)Cholera (D) All of these

    ANS:(D) All of these

16. Chromosome number in human being(male) is
    (A) 44+xx (B) 44++xy (C) 46+xy (D) 46+xx

    ANS:(B) 44++xy

17. Acetabularia is a type of -
    (A) Bacteria (B) Algae
    (C) Protozoa (D) Single cell protein

    ANS:(B) Algae

18. Cleistogamous flowers are always-
    (A) Self-Pollinated (B) Cross-Pollinated
    (C) Both (D) None

    ANS:(A)Self-Pollinated

19. Flow of energy in food chain of an ecosystem is
    (A) Unidirectional (B) Bidirectional
    (C) Multidirectional (D) none

    ANS:(A) Unidirectional

20. Which Blood group is called universal donor?
    (A) B (B) A
    (C) AB (D) O

    ANS:(D) O

21. Which one is phenotypic ratio od Monohybrid cross?
    (A) 1 : 2 : 1 (B) 3 : 1
    (C) 9 : 3 : 3 : 1 (D) none

    ANS:(B) 3 : 1

22. Rhino Sanctuary is located in which state?
    (A) Assam (B) West Bengal
    (C) Uttar Pradesh (D) Bihar

ANS:(A) Assam

23. Formation of mRNA from DNA is called-
(A) Transcription (B) Replication
(C) Translation (D) None
ANS:(C) Translation

24. The safe noise level fixed by WHO is-
(A) 20-30 Decibel (B) 40 Decibel
(C) 75 Decibel (D) 90 Decibel
ANS:(A) 20-30 Decibel

25. Malaria is caused by-
(A) Male Culex (B) Female Anopheles
(C) Female Aedes (D) Males Anopheles
ANS:(B) Female Anopheles

26. Bio fertilizer is present in root nodules of which non-leguminous plant?
(A) Azotobactor (B) Clostridium
(C) Frankia (D) None
ANS:(A) Azotobactor

27. Which bacterium is used in formation of curd from milk?
(A) Clostridium (B) Lactobacillus
(C) Both (A) and (B) (D) Streptococcus
ANS:(B) Lactobacillus

28. Inheritance of acquired characters was given by-
(A) Darwin (B) Lamarck
(C) De. vries (D) Haeckel
ANS:(B) Lamarck

Section-II : (Non-Objective Type Question)

Question Nos. 1 to 11 are of short answer type. Each question carries 2 marks.
11x2=22

Short Answer Type Questions

1. **Explain the structure of DNA.**
ANS: DNA is found in all animal cells. In eukaryotes it is usually found with protein as nucleo protein but in some prokaryotes it occurs independently. DNA is also present in mitochondria.
Structure: DNA is considered as the hereditary material which carries the genes. It is a large molecule built out of smaller molecules called monomer units or nucleotides. The monomer nucleotide of DNA has two parts a backbone and side chain. The backbone consists of a sugar molecule; deoxyribose and a phosphate group. The backbone is a long chain of sugar-phosphate and attached to each sugar residue is a side chain of a base. The bases are of four different kinds. A purine base-adenine or guanine. A pyrimidine base-cytosine or thymine.
According to Watson and Crick(a953) the structure of DNA is as follows:-

(i) DNA molecule consists of two long strands of nucleotides which are spirally coiled around a central axis. Due to this coiling arrangement deep and shallow grooves are formed.
(ii) The length of the one turn is 34A and in each turn there are 10nucleotides. So the distance between successive nucleotides in any chain is 3.4A, the width of DNA molecule is 20A.
(iii) In each strand deoxyribose sugar and phosphate groups are arranged alternately,
(iv) Either a purine or pyrimidine base is attached with deoxyribose.
(v) Purine base of one strand is connected with pyrimidine base of the other by hydrogen bonds.
(vi) Adenine always pairs with thymine(A-T) and guanine with cytosine (G-C)

2. **What is global warming? Explain its effect and ways to overcome it.**
ANS: If there is excess increase in concentration of greenhouse gases in the atmosphere, then it would retain more and more of the infrared radiation, resulting in enhanced greenhouse effect. Consequently, there will be increase in the global mean temperature which is referred as global warming.
Effects:- (a) It will cause melting of polar ice cap and glaciers.
(b) It will also cause rise in sea level.
(C) The climatic conditions will change and it will increase threat to human health.

3. **What is test tube baby?**

ANS: The fusion of ovum and sperm is done outside the body of woman to forma zygote which is allowed to divide to form embryo. Thus embryo is then implanted in uterus where it develops into a fetus which in turn develop into child. This is called test tube baby.

4. **Write any four sexually transmitted diseases with their causative pathogens**
ANS: Sexually transmitted disease are as follows:
(i) Gonorrhea – Neisseria gonorrheal
(ii) Syphilis – Treponema pallidum.
(iii) Genital warts – Human papilloma virus.
(iv) AIDS = Human immune virus (HIV)

5. **Differentiate between spermatogenesis and Oogenesis.**
ANS:

Spermatogenesis	Oogenesis
(i) It occurs in the testis	(i) It occurs in ovary
(ii) No polar body is formed	(ii) Polar bodies are formed
(iii) Spermatogoriam forms spermatozoa	(iii) Oogonium forms one ovum
(iv) Sperms are motile	(iv) Ova are not motile

6. **Write the function of blood**
ANS: Followings are the main functions of the blood:
(i) **Transportation of nutrition:** Blood carries digested food, water chemical substances, hormones and vitamins to the tissues.
(ii) **Disposal of carbon dioxide:** Active tissues continuously liberate carbon dioxide as a result of oxidation. The blood plasma and erythrocytes pick up CO_2 and take it to the organs of breathing from where it is released.
(iii) **Transport of oxygen:** It transports oxygen into the tissues for oxidation of food stuffs. Oxygen is located in the red blood corpuscles and released in the tissues.
(iv) **Regulation of body temperature:** Where as muscular activity produces heat, the lungs lose heat with expired air. Blood as circulating fluid carries heat from the regions where it is produced to other parts from where heat is lost. Thus it regulates body temperature.

7. **Write the ecological adaptations among xerophytic plants.**
ANS: Xerophytic adaptations are: (i) Presence of thick and waxy cuticle. (ii) Presence of sunken stomata and stomatal hairs.

8. **What is DNA probe? Give its application in Biotechnology.**
ANS: DNA probe is a technique to identify a segment of DNA using a known sequence of nucleotide bares from a DNA strand to detect a complementary sequence in the

samples. DNA probe is used for identification and isolation of DNA and diagnosis of same diseases.

9. **Define Biodiversity and explain its importance.**
ANS: The diversity of animal species and plant species in a region is known as biodiversity.
Importance: Biodiversity has prime role in agriculture as a source of new crops, material for breeding improved varieties and new biodegradable pesticide. The wheat, corn, and rice and the three main major carbohydrate crops which yield two third of the food sustaining the human population. The commercial domestic species are crossbreed with their wild relatives to improve their traits. The genes of wild species are used to confer new properties such as diseases resistance to improve yield indomesticated species e.g. rice grown in Asia is protected from diseases by genes received from a single wild rice species (Oryza nivara) from India.

10. **What is sterilization? Explain methods of sterilization.**
ANS Sterilization is a method of birth control-
(i) **Male sterilization:** The accepted methods for sterilization male is called vasectomy. It is surgical method in which each vas deferens is cut just above the epididymis.
(ii) **Tubectomy:** In female the fallopian tubes are cut.

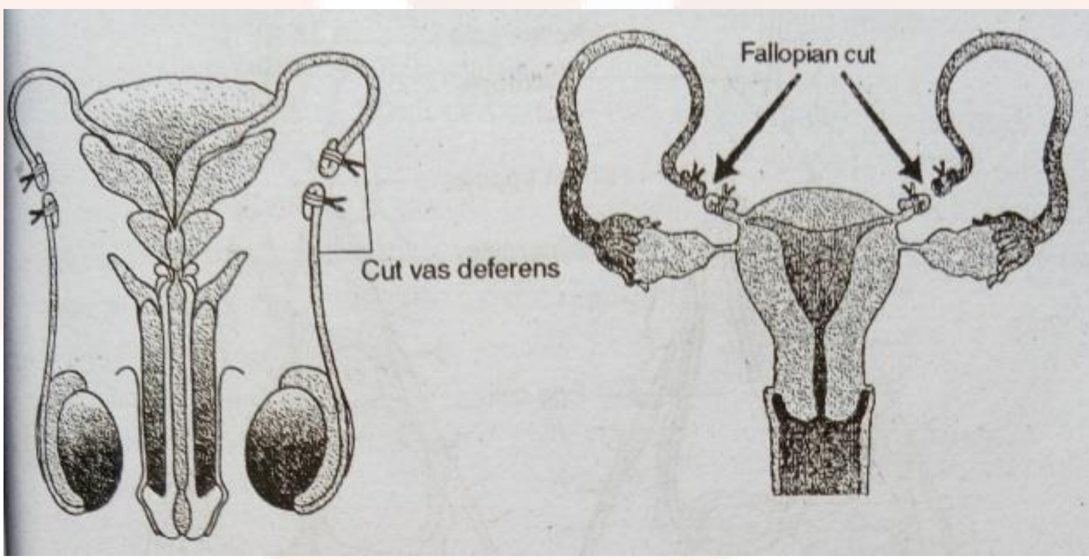


Fig. (a) Male Sterilization **Fig. (b) Female Sterlization**

11. **Define the following:**
    a. **Endoparasitic protozoans**
    b. **Ovule**

ANS: **(a) Endoparasitic protozoans:** Some protozoans line inside the body organism is known as protozoan endoparasites. E.g. Entomoeba histolytica, plasmoedium vivase etc.

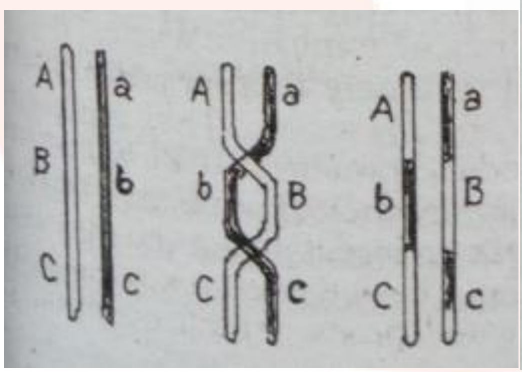
(B) Ovule: Ovule is an integumented megaporangium found in spermatophytes which develops into seed after fertilization.

Question Nos. 12 to 15 are of long answer type. Each question carries 5 marks

4x5=20

Long Answer Type Questions

12. Define crossing over. Describe the process with the help of diagrams.
Or,What do you understand by double fertilization? Explain it with the help of diagrams.
Q12: Define crossing over. Describe the process with the help of diagrams.
ANS: Crossing over is a recombination of genes due to exchange of genetic material between two homologous chromosomes. It is the mutual exchange of segments of genetic material between nonsister chromatids of two horologers chromosomes. So as to produce recombination or new combinations of genes.
The nonsister chromatids in which exchange of segment has occurred are called recombination or cross overs while the other chromatids in which crossing over has not taken place are known as parental chromatids or non-cross-overs.
Mechanism of crossing over: Chromosomes get replicated in S-phase of interphase. Therefore, laptotene chromosomes are double stranded though the two strands are not visible due to presence of nucleoprotein complex in between the chromatids.



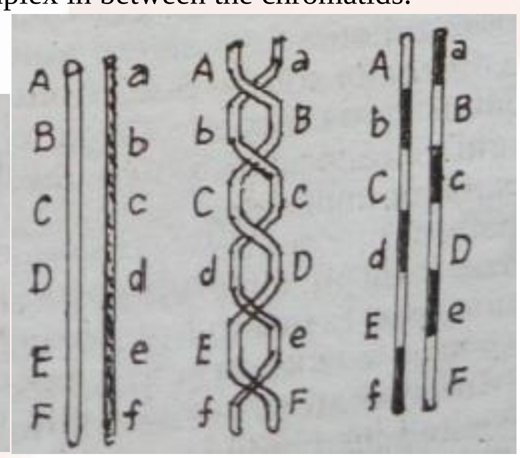


Fig: Double crossing over **Fig: Multiple crossing over**

(i) First of all homologous chromosomes pair. (ii) Each member of pair divide into two chromatids. Thus each monovalent chromosome become bivalent. (iii) If crossing over takes place then one chromosome twist around the corres ponding chromatid of the other chromosome. (iv) At this point break occurs and broken par of one chromatid join the other broken part of other corresponding chromatid resulting in exchange of chromatid pair containing the genes. (v) The point at which crossing over takes place called chiasmata. Now each centromere of the homologous chromosome start to separate from each other and the point of crossing over shifts towards the head chromosome. This movement of chiasmata is called terminalization. At the end of this chromatids get separated.

Q12: Or, What do you understand by double fertilization? Explain it with the help of diagrams.

ANS: The male gamete to emerges outside after the entry of embryo sac into pollen tube only one male gamete of embryo sac fuses with egg cell to form zygote while the other gamete fuses with polar nuclei. Both polar nuclei before fertilization form secondary nucleus. The structure formed by the fusion of secondary nucleus and male gamete is called Primary endosperm nucleus. In this way the process of fertilization completes.

Thus, in the event of fertilization, the function of male gamete with egg cell to form diploid zygote and the fusion of secondary nucleus with male gamete is call Double fertilization. Double fertilization takes place angiospermic plants only.

Significance of Double fertilization:

(i) In angiospermic plants there is special significance of double fertilization. If there is synamy only in plants and not the triple fusion, then no formation of endosperm & it will result into the formation of undeveloped embryonated or seed without embryo.

(ii) Endosperm formation is the result of double fertilization only. Thus, endosperm provides nutrients to embryo. There is maternal and paternal chromosome in endosperm.

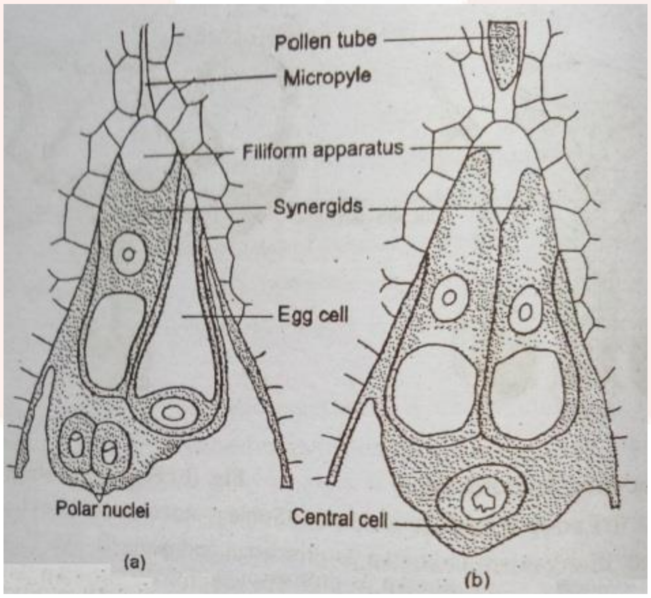


Fig: Fertilization (a) Mature embryo Sac (b) Entry of Pollen tube through ovule

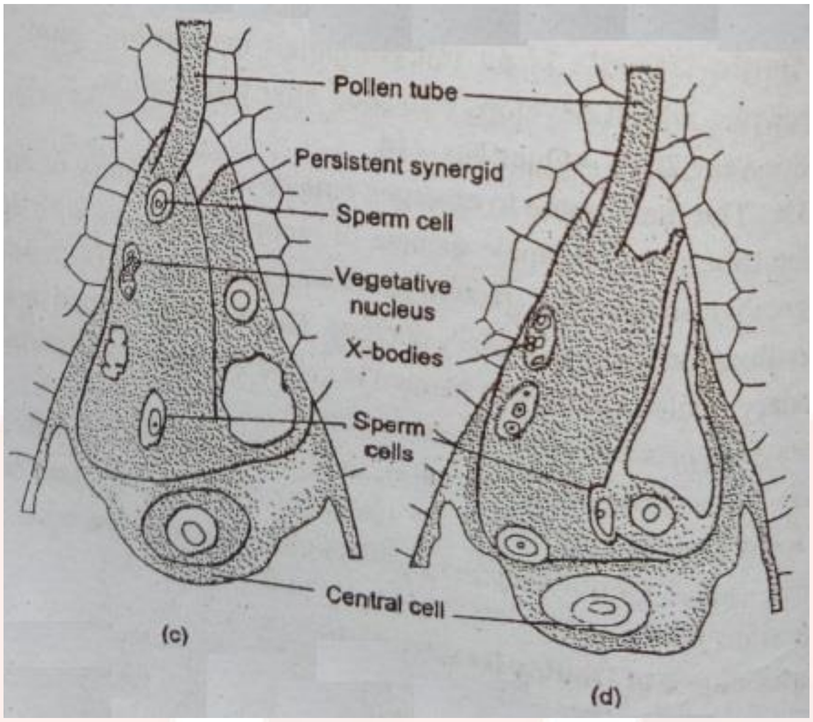
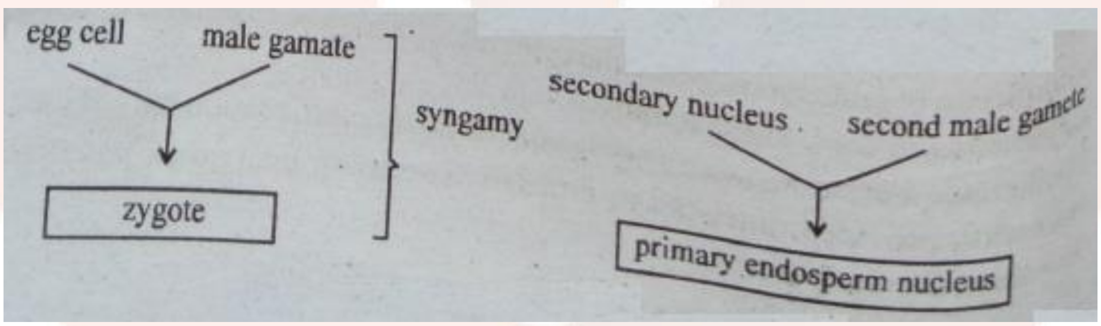


Fig: (C) A diploid cell resulting from the fusion of two haploid gametes
(d) Fusion Double fertilization of two haploid gametes



13. Giving suitable examples, describe the process of sex-determination.
Or, With the help of suitable sketches, describe the process of Spermatogenesis.
Q13: Giving suitable examples, describe the process of sex-determination.
ANS: In human being, male is heterogametic and thus posses AA & XY while female is homogametic & has AA and XX.
The male gametes produced are of two types. One containing A +X & other containing A+Y chromosome (1:1 ratio) whereas female gamete is always A+X. When male (A+X) fertilizes an (A+X), 2A and XX individual is produced which is daughter. On the contrary, if the A+Y male gamete fertilizes an egg (A+X) the result is a son-
AA+XX X AA+XY

Mother		Father	
Gamete A+X egg	A+X		A+Y sperms
AA+XX	AA+XY		
Daughter	Son		

Fig: Sex determination in human being at the time of fertilization

Q13: Or, With the help of suitable sketches, describe the process of Spermatogenesis.

ANS: Spermatogenesis is the process of sperm formation that takes place inside testis. This takes place in various stages as such-

(a) Multiplication Phase: Whenformation of large number of spermatogonia from germinal epithelium takes place.

(b) **Growth Phase:** During which growth of spermatogonia takes place into primary spermatocytes

(c) **Maturation Phase:** During which formation of secondary spermatocyte, spermatids and finally sperms takes place.

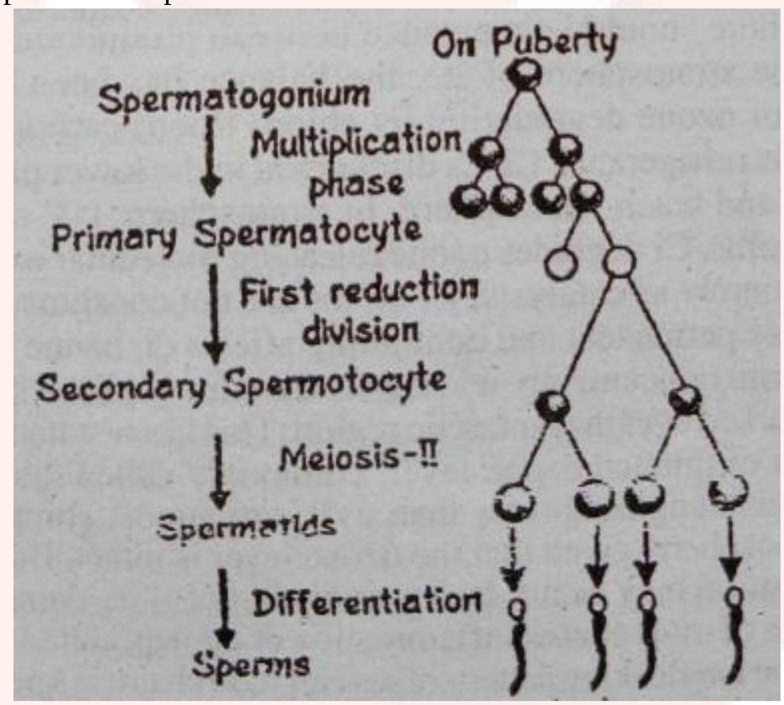


Fig: Spermatogenesis

14. Define Interaction of Genes with suitable crossing and examples.

Or, What are the basic principles of Immunology and their application?

Q14: Define Interaction of Genes with suitable crossing and examples.

ANS:A sort of gene interaction is seen when one gene modifies or mash the action of other gene. This is known as Epistasis, and it can give rise to unusual ratio in genetic cross. The gene which can over ride the expression of other, is epistatic gene and the other which get masked is the hypostatic gene. Dominant and recessive alleles, which by definition occur at the same loci on homologer chromosome.

Name of different types of	Modifies PR ratio	Example

Epistasis		
(i) Dominant epistasis	9 + 3 (12) : 3 : 1	Fruit colour in summer squash
(ii) Duplicate dominant epistasis	9 + 3 + 3 (15) : 1	Fruit shape in shepherd purse
(iii) Recessive epistasis	9 : 3 : (4) 3 + 1	Coat colour of mice
(iv) Duplicate Recessive epistasis	9 : (7) 3 + 3 + 1	Lathyrus oderatus
(v) Gene with cumulative effect	9 : (6) 3 + 3 : 1	Fruit shape in summer squash
(vi) Dominant Vs recessive epistasis	13 : 3	Feather colour of fowl

Q14 Or, What are the basic principles of Immunology and their application?

ANS: The resistance of the body to occurrence of any disease is know as immunity. Study of the ability of an organism to resist a disease its called immunology.

There are two main types of immunity:

(i) **Innate Immunity:** This type of immunity is inherited by the organisms from their parents.

(ii) **Acquired immunity:** This immunity is acquired in life time when organisms produces antibodies.

15. What is Ozone layer? Write its effects on atmosphere.

Or, What is food web? Explain it with the help of suitable examples.

Q15: What is Ozone layer? Write its effects on atmosphere.

ANS: Ozone gas is continuously formed by the action UV rays on molecular oxygen and also degraded into molecular oxygen in the stratosphere. There should be a balance between production and degradation of ozone in the stratosphere of lte, the balance has been disrupted due to enhancement of ozone degradation by chloro fluoro carbons (CFCs). CFCs find wide use as refrigerants. CFCs discharged in the lower part of the atmosphere move upward and reach stratosphere. In stratosphere, UV rays acts on them releasing CI atoms. CI degrades ozone releasing molecular oxygen, with these atoms acting merely as catalysts, CI atoms are not consumed in the reaction. Hence, there are permanent and continuing affects on ozone levels. Although ozone depletion is occurring widely in the stratosphere , the depletion is particularly marked over the Antarctic region. This has resulted in the formation of a large area of thinned ozone layer, commonly called ozone hole. UV radiation of wavelengths shorter than uvB, are almost completely absorbed by earth’s atmosphere, given that the ozone layer is intact, But uv-B damages DNA and mutation may occur. It causes aging skin, damage of skin cells and a high dose of uv-B causes inflammation of cornea, called snow blindness.

Q15: Or, What is food web? Explain it with the help of suitable examples.

ANS: The interlocking pattern of several food chains is known as food web. Thus food web may be defined as the relationship in which a predator eats several types of food and every kind of food is eaten by many different organisms and is the outcome of the interaction between different types of food chains.

e.g. Man and other organism are at the same time herbivore and carnivore and hence occupy different tropic levels in the food chain.